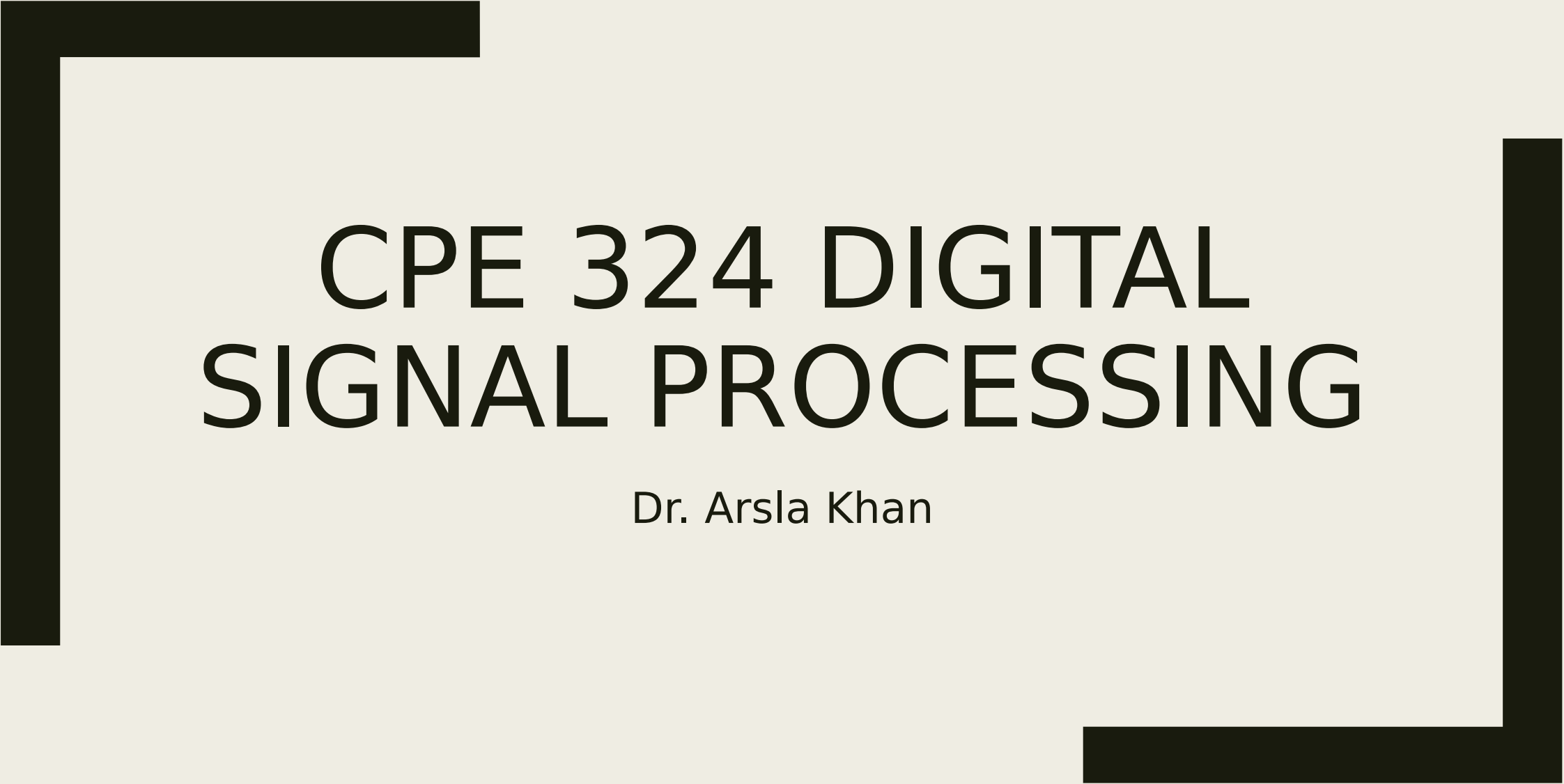


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CPE 324 DIGITAL SIGNAL PROCESSING

Dr. Arsla Khan

Course Overview

Topics and Milestones:

- Some basic concept in digital signal processing
- Digital Signals in Time and Frequency Domains
- Discrete Time LTI Systems
- Z-Transform
- Frequency domain analysis of signals and LTI systems

Sessional I Exam

- Sampling, Quantization of Signals, A/D and D/A Signal Transformation
- Discrete Fourier Transform and Fast Fourier Transform

Sessional II Exam

- Discrete Time System Implementation
- Digital Filters: FIR, IIR

Terminal Exam

Course Information

Textbook:

John G. Proakis and Dimitris G. Manolakis, *Digital Signal Processing: Principles, Algorithms, and Applications*, 4th edition, 2007

Reference Books:

- **Discrete-Time Signal Processing** by Alan V. Oppenheim, Ronald. W. Schaffer, Pearson Education (OS)
- **Digital Signal Processing: A Computer-based approach** by S. K. Mitra, McGraw-Hill Science (SKM)
- **Schaums Outline of Digital Signal Processing**, 2nd Edition (Schaum's Outline Series) by Monson Hayes

What is *Digital Signal Processing*

Signal: A parameter (electrical quantity or effect) that can be varied in such a way as to convey some information

Mathematically, a signal is a **function**. A **function** is a dependent variable of some variables which are independent variables.

A signal can be represented as a function $x(t)$ and consists of:

1. One or more **dependent variable** components (e.g., air pressure x , R-G-B color $[x_1 \ x_2 \ x_3]^T$).
2. One or more **independent variables** (e.g. time t , 3-D spatial location represented by $(s_1 \ s_2 \ s_3)$).

Independent variable: Do not limit yourself to TIME only !

What is *Digital Signal Processing*

Signal: A parameter (electrical quantity or effect) that can be varied in such a way as to convey some information

Examples:

- Pressure as a function of altitude, sound as a function of time, color as a function of space, . . .
- $x(t) = A \cos(\omega t + \Phi)$, $x(n, m) = (n + m)^2$,
 $x(t) = 4\sqrt{t} + t^3$ etc.

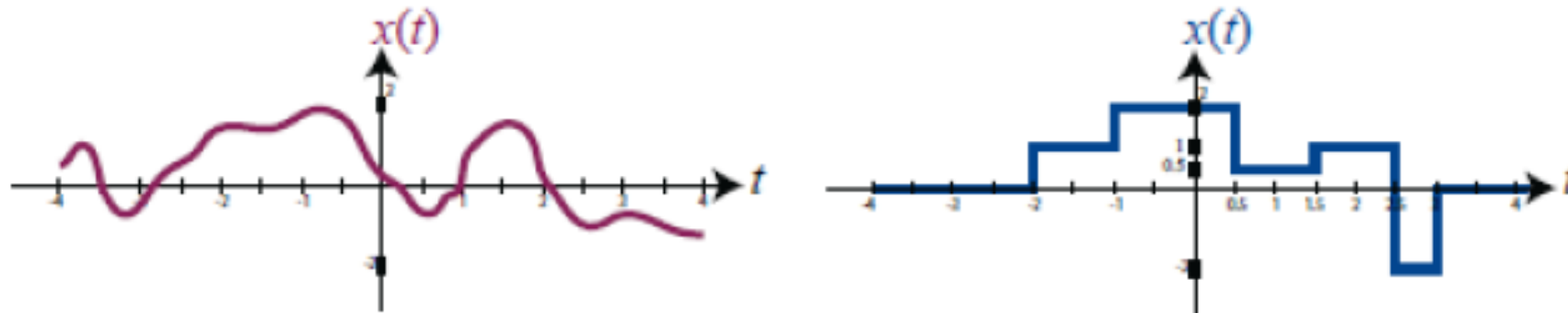
Note: We will use time to represent independent variable in this course. In general, it could be any other independent variable

What is *Digital Signal Processing*

Continuous Time vs. Discrete Time Signals:

Continuous-Time Signals: Signal is defined for every value of time in a given interval (a, b) where $a \geq -\infty$ and $b \leq \infty$

Examples: Voltage as a function of time, Pressure as a function of height etc.



What is *Digital Signal Processing*

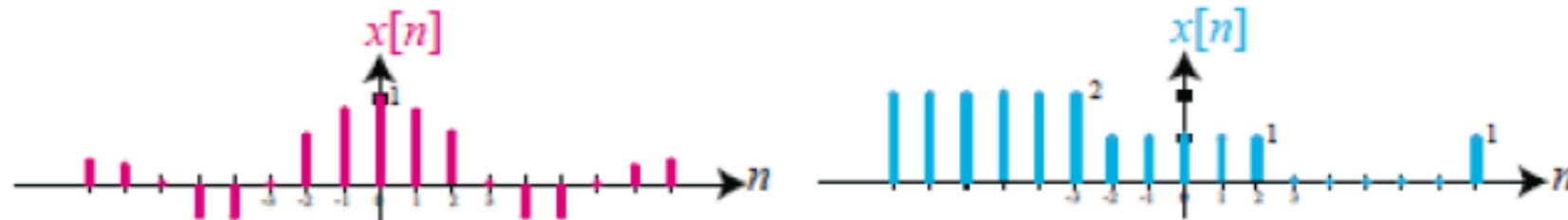
Continuous Time vs. Discrete Time Signals:

Discrete-Time Signals:

Signal is defined only for certain specific values of time; typically taken to be equally spaced points in an interval in a given interval (a, b) where $a \geq -\infty$ and $b \leq \infty$

Examples:

Number of stocks traded per day, Average income per province.



Thank You !!!

Chapter 2: Overview (Revision of SNS)

Classification of Signals

- Based upon their characteristics and nature of availability, the signals can be classified

- *Continuous Time Signal*
- *Discrete Time Signal*
- *Even Signal*
- *Odd Signal*
- *Periodic Signal*
- *Aperiodic Signal*
- *Energy Signal*
- *Power Signal*
- *Deterministic Signal*
- *Random Signal*

Exp : For the signal given, sketch $x[2n]$ and $x[n/2]$

$x[2n] \rightarrow$ Down-sampling

- $2n = -2 \rightarrow n = -1$
- $2n = -1 \rightarrow n = -1/2 = -0.5$
- $2n = 0 \rightarrow n = 0$
- $2n = 1 \rightarrow n = 1/2 = 0.5$
- $2n = 2 \rightarrow n = 2/2 = 1$
- $2n = 3 \rightarrow n = 3/2 = 1.5$
- $2n = 4 \rightarrow n = 4/2 = 2$

$$\frac{-1}{2} \quad \frac{0}{2} \quad \frac{1}{2} \quad \frac{2}{2} \quad \frac{3}{2} \quad \frac{4}{2}$$

- Thus, for $x[2n]$, pick every 2nd sample and discard other samples
- Similarly for $x[3n]$, pick every 3rd sample and discard other samples

$x[n/2] \rightarrow$ Up-sampling

- $n/2 = -2 \rightarrow n = -4$
 - $n/2 = -1 \rightarrow n = -1 \times 2 = -2$
 - $n/2 = 0 \rightarrow n = 0$
 - $n/2 = 1 \rightarrow n = 1 \times 2 = 2$
 - $n/2 = 2 \rightarrow n = 2 \times 2 = 4$
 - $n/2 = 3 \rightarrow n = 3 \times 2 = 6$
 - $n/2 = 4 \rightarrow n = 4 \times 2 = 8$
-
- Thus, $x[n/2] \rightarrow$ place samples at every 2nd place whereas zeros will be placed in between the samples
 - Thus, $x[n/3] \rightarrow$ place samples at every 3rd place whereas zeros will be placed in between the samples

Combined Operations

General Rule for Combined Operations

- First, do the time shifting operation
- Then, do the time scaling operation
- Then perform time folding

Exp : For the signal given, sketch $x(-2t + 3)$

$$x(t+3)$$

$$x(2t+3)$$

$$x(-2t+3)$$

Thank You !!!